

ABHINAV LAHARIYA

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Professional Summary

Machine Learning Engineer with almost 5 years of experience building and deploying ML systems for security threat detection at scale. Strong background in classical machine learning and deep learning, with hands-on experience in data-driven modeling, feature engineering, model optimization, and production integration.

Experience

Machine Learning Engineer

Intel

July 2021 - Present

Bangalore, India

1. Targeted Detector – Threat Detection Technology

- Designed, trained, and deployed ML models for real-time detection of cyber threats using CPU telemetry .
- Built scalable feature engineering pipelines to transform high-volume telemetry data into model-ready signals, improving detection robustness.
- Performed extensive model evaluation and tuning using ROC-AUC, Precision-Recall, F1-score, confusion matrix analysis, and threshold/false-positive optimization to balance detection accuracy and reliability.
- Optimized model performance through hyperparameter tuning and algorithm selection, leading to improved detection accuracy in production environments.
- Trained and evaluated models using AWS EC2, managed datasets on S3, and tracked experiments with DVC to ensure reproducible and well-organized research results.
- Worked closely with security researchers and platform teams to translate threat research into deployable ML solutions.

2. AI-Based Telemetry Selector

- Enhanced existing telemetry selection algorithms to better identify high-value features for security modeling, reducing data collection overhead and improving model efficiency.
- Applied continuous learning techniques to adapt telemetry selection strategies as new attack patterns emerged.
- Built data processing pipelines to collect, clean, and aggregate telemetry across multiple hardware platforms.
- Collaborated with cross-functional teams to integrate solutions into existing security frameworks.

Software Development Intern

Intel

Jan 2021 – Jun 2021

Bangalore, India

- Designed and implemented multiple Proofs of Concept (POCs) to evaluate algorithmic, architectural, and code-level optimizations for performance-critical systems.
- Benchmarked alternative approaches and identified performance bottlenecks, leading to measurable improvements in efficiency and system throughput.
- Collaborated closely with R&D Team to validate and integrate successful POCs into existing development workflows.
- Worked with production-grade codebases, gaining hands-on experience in version control, debugging, and performance analysis.

Projects

Real-time Emotion and Gender Classification

- Designed and implemented deep learning models for emotion and gender classification using image data.
- Explored and evaluated different model architectures and training strategies to improve emotion classification performance.
- Conducted systematic experiments and performance evaluations across multiple emotion categories to validate model behavior.
- Published the experimental findings in a research paper on arXiv: <https://arxiv.org/abs/2111.07746>.

Technical Skills

- Programming & CS Fundamentals: **Python, C++, Data Structures, Algorithms.**
- Machine Learning: **Supervised & Unsupervised Learning, Regression, Classification, Random Forests, Gradient Boosting, XGBoost, Genetic Algorithms.**
- Deep Learning: **Neural Networks, CNN, RNN, LSTM, Attention Mechanisms.**
- Transformers & NLP: **Transformer Architecture, BERT, Tokenization, Embeddings, TF-IDF, Word2Vec, Sentiment Analysis.**
- Model Evaluation & Optimization: **ROC-AUC, Precision-Recall, F1-score, Hyperparameter Tuning, Cross-Validation.**
- Tools & Frameworks: **PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Docker, AWS, DVC, Git, Jupyter, PyCharm.**

Education

Indian Institute Of Information Technology , Allahabad

July 2019 – June 2021

M.Tech in Machine Learning and Intelligent Systems

Prayagraj

Madhav Institute Of Technology & Science, Gwalior

July 2015 – June 2019

B.E in Computer Science & Engineering

Gwalior